

EE5750 Quantum Error Correction

Homework 3

Due: 20 Aug

As part of the Homework assessment provide solutions to only the questions marked as (***Homework***). The rest of the questions are practice problems.

1. (***Homework***) Consider the Hilbert space $\mathbb{C}^d$, and let $\{|\psi_i\rangle : i = 0, \dots, d-1\}$ be any orthonormal basis (this need not be the standard basis). Let $P_i = |\psi_i\rangle\langle\psi_i|$.
 - (a) Check if P_i is a projection operator.
 - (b) Find the rank and the column space of P_i .
 - (c) Verify $P_i P_j = 0$ if $i \neq j$.
 - (d) Verify completeness condition.

Remark: When we choose $|\psi_i\rangle = |i\rangle$, i.e., when the orthonormal basis used for measurement is the standard basis, we say that we ‘perform measurement in the computational basis.’

2. (***Homework***) Let P and Q be projection operators such that $PQ = 0$.
 - (a) Find QP .
 - (b) If $|\psi\rangle$ is an eigenvector of Q with eigenvalue 1, then find the value of $P|\psi\rangle$.
 - (c) Is $P + Q$ a projection operator?

Remark: In this case, one can show that $\text{rank}(P + Q) = \text{rank}(P) + \text{rank}(Q)$, and $P + Q$ projects onto the subspace obtained as the ‘sum’ of the column spaces of P and Q .

3. If P is a projection operator, is $I - P$ a projection operator too? Are the column spaces of P and $I - P$ orthogonal to each other?
4. (***Homework***) Consider classical measurement of classical states as done in Lecture 6, i.e., consider a partition $S_1, \dots, S_L$ of $\{0, 1, \dots, d-1\}$ and distinct real numbers $\lambda_1, \dots, \lambda_L$. The measurement device outputs λ_ℓ if the classical random variable X stored in the classical register belongs to S_ℓ , i.e., $X \in S_\ell$.

Define an operator $M = \sum_{\ell=1}^L \lambda_\ell P_\ell$, where $P_\ell = \sum_{j \in S_\ell} |j\rangle\langle j|$.

- (a) What does the value $\text{Tr}(M\rho)$ denote/signify, where ρ is a classical state?
- (b) What about $\text{Tr}(M^2\rho)$?

Remark: Note that M is a Hermitian operator and it contains all the information about $P_1, P_2, \dots$ and $\lambda_1, \lambda_2, \dots$, that is, given the matrix M we can find all P_i and λ_i , this is how:

- $\lambda_1, \dots, \lambda_L$ are the distinct eigenvalues of M (that is, all eigenvalues not counting repetitions).
- P_ℓ is the projection onto the eigenspace corresponding to the eigenvalue λ_ℓ . In particular, if the multiplicity of the eigenvalue λ_ℓ is m_ℓ , then there will be m_ℓ orthonormal eigenvectors of M with eigenvalue λ_ℓ , say $|\psi_{\ell,1}\rangle, \dots, |\psi_{\ell,m_\ell}\rangle$, then

$$P_\ell = |\psi_{\ell,1}\rangle\langle\psi_{\ell,1}| + \dots + |\psi_{\ell,m_\ell}\rangle\langle\psi_{\ell,m_\ell}|$$

Such a matrix M , in the context of measurement, is called an *observable*.

5. **(*Homework*)** This question is based on the remark of the last question (about observables). The Pauli Z operator is defined as

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

- (a) Find all the eigenvalues and eigenvectors of Z .
 - (b) If Z is used as an observable for a measurement, find the corresponding measurement outputs $\lambda_1, \lambda_2, \dots$ and the projection operators $P_1, P_2, \dots$.
 - (c) Do the same for the observable $Z \otimes Z$.
6. Let $C = A \otimes B$. Verify the following.
- (a) A and B are unitary $\Rightarrow C$ is unitary.
 - (b) A and B are Hermitian $\Rightarrow C$ is Hermitian.
 - (c) A and B are skew-Hermitian $\Rightarrow C$ is Hermitian.

7. *Unitary Diagonalizability.*

We say that a square matrix A is *unitarily diagonalizable* if there exists a unitary matrix U and a diagonal matrix Λ such that

$$A = U \Lambda U^\dagger.$$

The characterization of unitarily diagonalizable matrices is well known:

$$A \text{ is unitarily diagonalizable if and only if } AA^\dagger = A^\dagger A.$$

- (a) Verify if the following matrices are unitarily diagonalizable: Hermitian, skew-Hermitian, unitary.
- (b) Verify that the columns of U are the eigenvectors of A and the diagonal entries of Λ are the eigenvalues.

Hint: Use the fact $AU = U\Lambda$.

Remark: A is unitarily diagonalizable if and only if it has a complete orthonormal set of eigenvectors, i.e., $|\psi_1\rangle, \dots, |\psi_d\rangle$ are an orthonormal basis of $\mathbb{C}^d$ and each $|\psi_i\rangle$ is an eigenvector of A . These $|\psi_i\rangle$ are precisely the columns of U .

8. **(*Homework*)** Suppose A and B are unitarily diagonalizable, verify if $A \otimes B$ is unitarily diagonalizable. Find the eigenvalues and eigenvectors of $A \otimes B$ based on the eigenvalues and eigenvectors of A and B .
9. **(*Homework*)** *Kraus form of partial trace.*

We consider applying the partial trace operation on an arbitrary $d_1 d_2 \times d_1 d_2$ matrix R (that is, R need not be a density operator).

- (a) Verify the following Kraus form

$$\text{Tr}_B(R) = \sum_{j=0}^{d_2-1} (I_{d_1} \otimes \langle j|) R (I_{d_1} \otimes |j\rangle).$$

You must check the completeness condition as well.

Remark: This expression is sometimes written in shorthand as

$$\sum_{j=0}^{d_2-1} \langle j|_B R |j\rangle_B$$

(b) Find the Kraus form for tracing out A .

10. (***Homework***) The *Pauli operators* are defined as follows

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

- (a) Verify that every 2×2 Hermitian operator is a linear combination of the Pauli operators with *real coefficients* used for the linear combination.
- (b) Is *every* 2×2 operator some complex linear combination of the Pauli operators? Do these operators form a basis for $\mathbb{C}^{2 \times 2}$?

HINT: You must check that every 2×2 matrix is some linear combination of I, X, Y, Z , and you must also show that if $\alpha_1 I + \alpha_2 X + \alpha_3 Y + \alpha_4 Z = 0$, then all $\alpha_i = 0$.

- (c) The n -qubit *Paulis* are the operators

$$\{A_1 \otimes \cdots \otimes A_n : A_i \in \{I, X, Y, Z\}, i = 1, \dots, n\}.$$

Argue that the n -qubit Paulis form a basis for $\mathbb{C}^{2^n \times 2^n}$.
